

Project 6 – Footprinting, Reconnaissance, and Scanning

Answer Sheet

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Date: 04/15/2026

Part 1 – Passive Reconnaissance / OSINT

Task 1 – Whois: Domain Identity & Ownership

Instructions: Use a public Whois lookup for testphp.vulnweb.com (or its parent domain) and record the requested information below.

Domain: vulnweb.com

Registrar: Gandi SAS

Organization / Owner: Invicti Security Limited

Creation Date: 2010-06-14

Expiration Date: 2027-06-14

Name Servers: ns-105-a.gandi.net ns-11-b.gandi.net ns-140-c.gandi.net

Notes: The domain vulnweb.com is registered with Gandi SAS and belongs to Invicti Security Limited. It's been around since 2010, so it's not a new domain. There are multiple name servers listed, which helps keep the site stable. Even though some contact info is private, there's still enough information available to give a general idea of who owns and manages the domain.

Task 1 Screen Shot

vulnweb.com

Updated 10 hours ago ↗

Domain Information	
Domain:	vulnweb.com
Registered On:	2010-06-14
Expires On:	2027-06-14
Updated On:	2025-11-17
Status:	client transfer prohibited
Name Servers:	ns-105-a.gandi.net ns-11-b.gandi.net ns-140-c.gandi.net
Registrar Information	
Registrar:	Gandi SAS
IANA ID:	81
Abuse Email:	abuse@support.gandi.net
Abuse Phone:	+33.170377661
Registrant Contact	
Organization:	Invicti Security Limited
Country:	MT
Email:	e7c92ba29faca2e997bfc9ce50d5c601-57140470@contact.gandi.net
Administrative Contact	
Email:	e7c92ba29faca2e997bfc9ce50d5c601-57140470@contact.gandi.net
Technical Contact	
Email:	e7c92ba29faca2e997bfc9ce50d5c601-57140470@contact.gandi.net

Task 2 – DNS Basics

Instructions: Use nslookup or dig for testphp.vulnweb.com and summarize the DNS information.

Command used: nslookup testphp.vulnweb.com

IP address(es) observed: 44.228.249.3

Record type(s) (A / CNAME / TXT, etc.): A record, TXT record

Additional notes: The DNS lookup shows the domain points to the IP address 44.228.249.3, so the server can be reached from the internet. There aren't any AAAA or CNAME records, which makes the setup look pretty straightforward. There is a TXT record for Google verification, meaning the site is connected to Google. Overall, this gives a general idea of how the domain is set up and what could be looked at next.

Task 2 Screen Shot

A records

IPv4 address	Revalidate in
> 44.228.249.3	3h

AAAA records

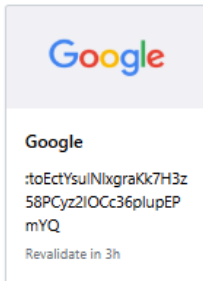
No AAAA records found.

CNAME record

No CNAME record found.

TXT records

Site ownership verification



NS records

No NS records found.

The name servers for this domain are inherited from one of its ancestor domains. Try its parent domain: vulnweb.com..

MX records

No mail servers found.

Other records

SOA

No SOA records found.

Look up DNS records for another domain name

With [DNS lookup](#), you can find the DNS record for any domain name or subdomain. When you enter the domain, it will show all the DNS records that are configured for that domain.

Task 3 – Hosting & Technology Stack

Instructions: Use Netcraft Site Report or a similar public site-technology report for testphp.vulnweb.com.

Tool used: Netcraft Site Report

<https://sitereport.netcraft.com/?url=http://testphp.vulnweb.com>

Hosting provider / ASN: Amazon - US West (Oregon) datacenter // [AS16509](#)

Country / Region: United States Oregon

Web server / tech stack reported: A specific web server (like Apache or nginx) was **not listed in the report**.

Other observations:

The site looks like it's hosted on **Amazon (AWS)** in the **US West (Oregon)** region based on the IP and network info.

The report didn't show a specific web server, but it appears to be a **PHP-based site**, and no exact tech stack was listed.

One thing I noticed is the site isn't using **HTTPS**, so there's no encryption.

Background

Site title	Not Present	Date first seen	April 2017
Site rank	1439	Primary language	English
Description	Not Present		

Network

Site	http://testphp.vulnweb.com	Domain	vulnweb.com
Netblock Owner	Amazon.com, Inc.	Nameserver	ns1.gandi.net
Hosting company	Amazon - US West (Oregon) datacenter	Domain registrar	gandi.net
Hosting country	us	Nameserver organisation	whois.gandi.net
IPv4 address	44.228.249.3 (VirusTotal)	Organisation	Invicti Security Limited, REDACTED FOR PRIVACY, REDACTED FOR PRIVACY, Malta
IPv4 autonomous systems	AS16509	DNS admin	hostmaster@gandi.net
IPv6 address	Not Present	Top Level Domain	Commercial entities (.com)
IPv6 autonomous systems	Not Present	DNS Security Extensions	Enabled
Reverse DNS	ec2-44-228-249-3.us-west-2.compute.amazonaws.com		

IP delegation

IPv4 address (44.228.249.3)

IP range	Country	Name	Description
::ffff:0:0:0:0/96	United States	IANA-IPv4-MAPPED-ADDRESS	Internet Assigned Numbers Authority
44.0.0.0-44.255.255.255	United States	NET44	American Registry for Internet Numbers
44.192.0.0-44.255.255.255	United States	AMAZO-4	Amazon.com, Inc.
44.224.0.0-44.255.255.255	United States	AMAZO-ZPDIX	Amazon.com, Inc.
44.228.249.3	United States	AMAZO-ZPDIX	Amazon.com, Inc.

SSL/TLS

This is not a HTTPS site. If you're looking for SSL/TLS information try the [HTTPS site report](#).

Sender Policy Framework

A host's Sender Policy Framework (SPF) describes who can send mail on its behalf. This is done by publishing an SPF record containing a series of [rules](#). Each rule consists of a qualifier followed by a specification of which domains to apply this qualifier to. For more information please see [open-spf.org](#).

Warning: It appears that this host does not have an SPF record. There may be an SPF record on vulnweb.com: Check the [site report](#).

Setting up an SPF record helps prevent the delivery of forged emails from your domain. Please note that an SPF record will only protect the domain it is added to and not any [mail-enabled subdomains](#). It is recommended to add an SPF record to any subdomain with an MX record.

DMARC

DMARC (Domain-based Message Authentication, Reporting and Conformance) is a mechanism for domain owners to indicate how mail purporting to originate from their domain should be authenticated. It builds on SPF and DKIM, providing a method to set policy and to give reporting of failures. For more information please see [dmarc.org](#).

This host does not have a DMARC record. There may be a DMARC record on the site report for vulnweb.com: Check the [site report](#).

Web Trackers

Web Trackers are third-party resources loaded onto a webpage. Trackable resources include social sharing widgets, Javascript files, and images. These trackers can be used to monitor individual user behaviour across the web. Data derived from these trackers are primarily used for advertising or analytics purposes.

No known trackers were identified.

Site Technology (fetched 2 days ago)

Cloud & PaaS

Cloud computing is the use of computing resources (hardware and software) that are delivered as a service over a network (typically the Internet). Platform as a service (PaaS) is a category of cloud computing services that provide a computing platform and a solution stack as a service.

Task 4 – Google Dorks on testphp.vulnweb.com

Instructions: Run at least three Google queries using operators (site:, inurl:, filetype:, "quoted phrase"). For each query, record one interesting result and why it might matter.

Querysite:	Interesting Result (Title/URL)	Why this might matter (attacker or defender)
site: testphp.vulnweb.com	Index of /wvstests/pmwiki_2_1_19/ scripts/	This shows a directory listing, which means files are publicly accessible. An attacker could use this to find vulnerabilities, while a defender would see this as something that should probably be restricted.
site:testphp.vulnweb.com inurl:php	http://testphp.vulnweb.com/search.php	This shows a page that uses user input (search.php), which could be tested for things like SQL injection. From a security standpoint, it should be properly validated and secured.
site:testphp.vulnweb.com filetype:php	http://testphp.vulnweb.com/AJAX/categories.php	This shows a PHP file that could be handling data or requests. It could be tested for vulnerabilities, so it's something that should be secured.

Screenshot filenames (if any):

Query 1

The screenshot shows a Google search interface with the query 'site:testphp.vulnweb.com' entered in the search bar. The search results are displayed below the navigation bar, which includes links to 'AI Mode', 'All', 'Shopping', 'Images', 'Short videos', 'Forums', 'Videos', 'More', and 'Tools'. The first result is a 'Google promotion' for 'Try Google Search Console'. The second result is from 'VulnWeb' with the URL 'https://testphp.vulnweb.com' and the title 'Acunetix acuart - VulnWeb'. The third result is from '신라이앤씨' with the URL 'http://testphp.vulnweb.com/wvstests/scripts' and the title 'Index of /wvstests/pmwiki_2_1_19/scripts/'. The fourth result is from 'Acunetix VulnWeb' with the URL 'http://testphp.vulnweb.com/hpp' and the title 'HTTP Parameter Pollution Example'. The fifth result is from 'Acunetix VulnWeb' with the URL 'http://testphp.vulnweb.com/BuyProduct-2' and the title 'Thanks for buying Web Camera A4Tech PK-335E'. The sixth result is from 'Acunetix VulnWeb' with the URL 'http://testphp.vulnweb.com/RateProduct-2' and the title 'Thanks for rating Web Camera A4Tech PK-335E'. The seventh result is from 'Acunetix Web Vulnerability Scanner - Test websites' with the URL 'http://testphp.vulnweb.com/Mod_Rewrite_Shop/Details' and the title 'Laser Color Printer HP LaserJet M551dn, A4 ...'.

Google

site:testphp.vulnweb.com

AI Mode All Shopping Images Short videos Forums Videos More Tools

Google promotion

Try Google Search Console
www.google.com/webmasters/
Do you own **testphp.vulnweb.com**? Get indexing and ranking data from Google.

VulnWeb
https://testphp.vulnweb.com
Acunetix acuart - VulnWeb

신라이앤씨
http://testphp.vulnweb.com/wvstests/scripts
Index of /wvstests/pmwiki_2_1_19/scripts/
Index of /wvstests/pmwiki_2_1_19/scripts/ ../ version.php 11-May-2011 10:27 38. [Read more](#)

Acunetix VulnWeb
http://testphp.vulnweb.com/hpp
HTTP Parameter Pollution Example
check - Original article. [Read more](#)

Acunetix VulnWeb
http://testphp.vulnweb.com/BuyProduct-2
Thanks for buying Web Camera A4Tech PK-335E
Thanks for buying Web Camera A4Tech PK-335E. [Read more](#)

Acunetix VulnWeb
http://testphp.vulnweb.com/RateProduct-2
Thanks for rating Web Camera A4Tech PK-335E
Thanks for rating Web Camera A4Tech PK-335E. [Read more](#)

Acunetix Web Vulnerability Scanner - Test websites
http://testphp.vulnweb.com/Mod_Rewrite_Shop/Details
Laser Color Printer HP LaserJet M551dn, A4 ...

Query 2

Google

site:testphp.vulnweb.com inurl:php

AI Mode All Images Shopping Videos Forums News More Tools

 VulnWeb
http://testphp.vulnweb.com › search

<http://testphp.vulnweb.com/search.php>

 Acunetix Web Vulnerability Scanner - Test websites
http://testphp.vulnweb.com › listproducts

[pictures - Acunetix Art](#)

This is an example PHP application, which is intentionally vulnerable to web attacks. It is intended to help you test Acunetix. It also helps you understand how ... [Read more](#)

 VulnWeb
http://testphp.vulnweb.com › secured › newuser

[add new user](#)

ACUNETIX ART. [Read more](#)

 vulnweb.com
http://testphp.vulnweb.com › listproducts

[YouTube](#)

Share your videos with friends, family, and the world.

 VulnWeb
http://testphp.vulnweb.com › hpp

[HTTP Parameter Pollution Example](#)

check · link1 · link2. Original article. [Read more](#)

 Acunetix VulnWeb
http://testphp.vulnweb.com › hpp

Query 3

Google

site:testphp.vulnweb.com filetype:php

AI Mode All Shopping Images Short videos Forums Videos More Tools

Acunetix acuart
http://testphp.vulnweb.com › AJAX › categories XML
<http://testphp.vulnweb.com/AJAX/categories.php>

Acunetix VulnWeb
http://testphp.vulnweb.com › hpp
HTTP Parameter Pollution Example
check · Original article. [Read more](#)

vulnweb.com
http://testphp.vulnweb.com › redir › 28728.surkoph.ru
ODELICE Restaurants
主頁 我的資料 獎賞 禮品卡 預訂 分店位置 積分紀錄 到期日 消息通知 · logo · Home · Browse · Privacy Policy · Contact. English, Spanish. [Read more](#)

VulnWeb
http://testphp.vulnweb.com › hpp
HTTP Parameter Pollution Example
check · link1 · link2. Original article. [Read more](#)

vulnweb.com

Task 5 – Light OSINT on fullstackacademy.com (Google Only)

Instructions: Run at least three Google searches using site:fullstackacademy.com or its subdomains with at least one other operator. Record an example result and why it could matter from an OSINT or security perspective.

Query	Example Result (Title/URL)	Why this might matter
start.fullstackacademy.com	Start Fullstack Academy – start.fullstackacademy.com	This shows there's a separate subdomain used for starting or onboarding. From a security/OSINT perspective, subdomains can sometimes have login pages or different configurations that could be targeted if not secured properly.
site:start.fullstackacademy.com filetype:pdf	AWS Cloud Practitioner Essentials	This shows PDFs that are publicly accessible with course info. You can see what they teach and how things are set up, which could give someone insight into their systems or training.
site:fullstackacademy.com "syllabus"	USD Cyber Syllabus – start.fullstackacademy.com	This pulls up real syllabi, so you can see what they teach and what tools they use. That kind of info could help someone understand their setup or focus areas.

Screenshot filenames (if any):

Query 1

Google search results for `start.fullstackacademy.com`.

These are results for **start.fullstack academy**
Search instead for [start.fullstackacademy](#)

Sponsored results

- courses-si.professional.ucsb.edu**
<https://courses-si.professional.ucsb.edu/software/development/>
Full Stack Development Course | UCSB PaCE Online Course
Learn vite coding online, create a GitHub portfolio and get resume, interview support. Build AI-enabled web apps using ChatGPT, Copilot, and real labs.
- Full Stack With GenAI Course**
- Full Stack Program**
- Full Stack With GenAI Online**
- IBM Skill Modules**
- MERN Stack Skills**

- arapahoe.quickstart.com**
<https://arapahoe.quickstart.com/>
Online IT Bootcamps | Arapahoe CC IT Bootcamps
Break into IT in 10-16 weeks with flexible, fully online bootcamps built for beginners. Live instructors, hands-on labs, and career services to support your job search. Job-Ready Skills Training.
- Codecademy**
<https://www.codecademy.com/>

Query 2

Google search results for `site:start.fullstackacademy.com filetype:pdf`.

Fullstack Academy
<https://start.fullstackacademy.com/hubfs/WDF...> PDF

Eligibility Criteria
These requirements must be satisfied at the time of submitting the WDF Online Application. o. New York City resident: You live in one (1) of the five (5) ... [Read more](#)

Fullstack Academy
<https://start.fullstackacademy.com/hubfs/Documents>

AWS Technical Essentials
AWS Technical Essentials introduces you to essential AWS services and common solutions. The course covers the fundamental AWS concepts related to compute, ...

Fullstack Academy
<https://start.fullstackacademy.com/Documents> PDF

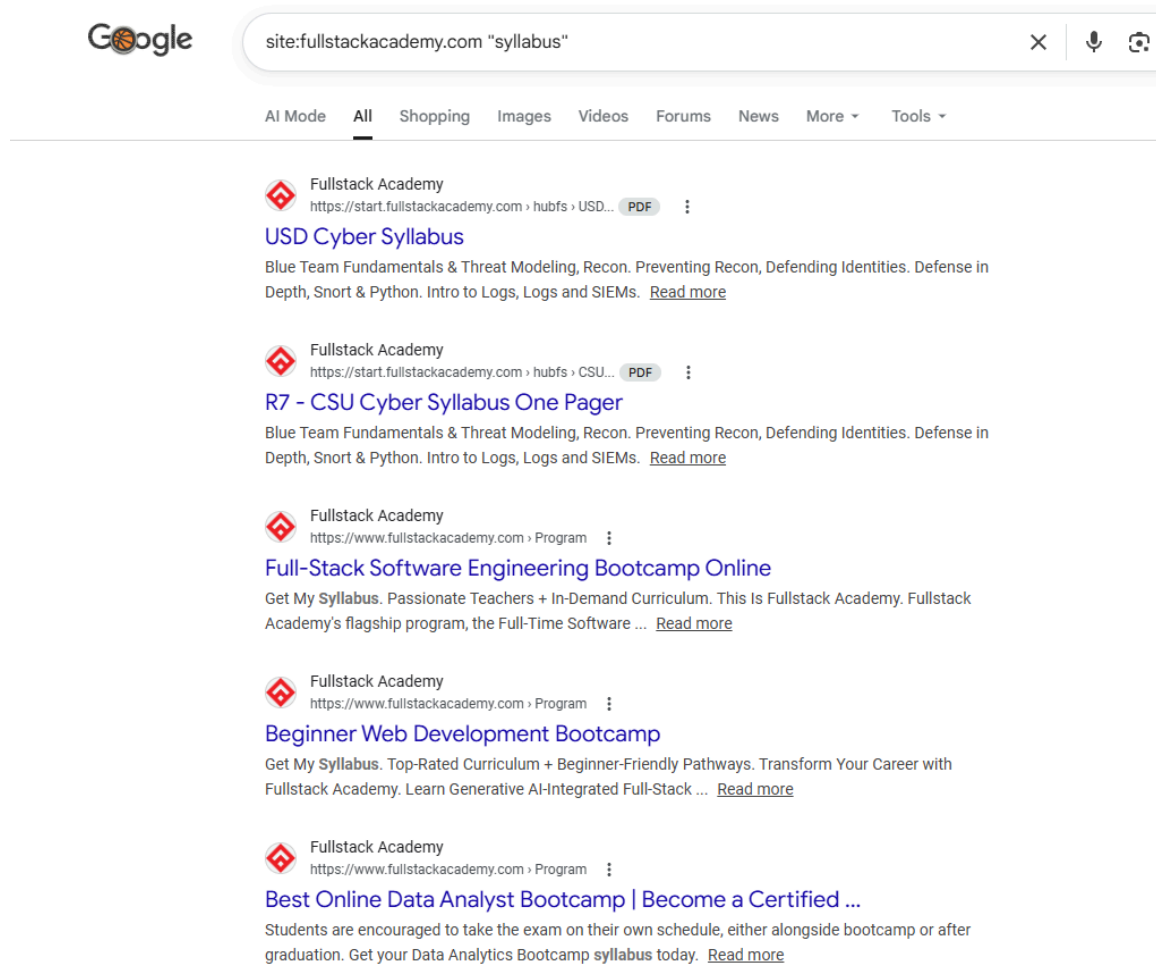
AWS Cloud Practitioner Essentials
This course is for individuals who seek an overall understanding of the Amazon Web Services (AWS) Cloud, independent of specific technical roles. [Read more](#)

Fullstack Academy
<https://start.fullstackacademy.com/Documents> PDF

CompTIA Security+ 601 certification
CompTIA Security+ 601 is a globally trusted certification that validates foundational, vendor-neutral IT security knowledge and skills.

Fullstack Academy
<https://start.fullstackacademy.com/hubfs/...> PDF

Query 3



The screenshot shows a Google search interface with the query 'site:fullstackacademy.com "syllabus"'. The search results are filtered by 'All' and show five entries from Fullstack Academy. Each entry includes the site logo, the URL, a PDF icon, and a title. The titles are: 'USD Cyber Syllabus', 'R7 - CSU Cyber Syllabus One Pager', 'Full-Stack Software Engineering Bootcamp Online', 'Beginner Web Development Bootcamp', and 'Best Online Data Analyst Bootcamp | Become a Certified ...'. Each entry also has a short description and a 'Read more' link.

Google

site:fullstackacademy.com "syllabus"

AI Mode All Shopping Images Videos Forums News More Tools

Fullstack Academy
https://start.fullstackacademy.com › hubfs › USD... PDF

USD Cyber Syllabus
Blue Team Fundamentals & Threat Modeling, Recon. Preventing Recon, Defending Identities. Defense in Depth, Snort & Python. Intro to Logs, Logs and SIEMs. [Read more](#)

Fullstack Academy
https://start.fullstackacademy.com › hubfs › CSU... PDF

R7 - CSU Cyber Syllabus One Pager
Blue Team Fundamentals & Threat Modeling, Recon. Preventing Recon, Defending Identities. Defense in Depth, Snort & Python. Intro to Logs, Logs and SIEMs. [Read more](#)

Fullstack Academy
https://www.fullstackacademy.com › Program

Full-Stack Software Engineering Bootcamp Online
Get My Syllabus. Passionate Teachers + In-Demand Curriculum. This Is Fullstack Academy. Fullstack Academy's flagship program, the Full-Time Software ... [Read more](#)

Fullstack Academy
https://www.fullstackacademy.com › Program

Beginner Web Development Bootcamp
Get My Syllabus. Top-Rated Curriculum + Beginner-Friendly Pathways. Transform Your Career with Fullstack Academy. Learn Generative AI-Integrated Full-Stack ... [Read more](#)

Fullstack Academy
https://www.fullstackacademy.com › Program

Best Online Data Analyst Bootcamp | Become a Certified ...
Students are encouraged to take the exam on their own schedule, either alongside bootcamp or after graduation. Get your Data Analytics Bootcamp syllabus today. [Read more](#)

Task 6 – Passive Recon Summary

Instructions: Summarize the overall exposure and key findings from your passive recon.

Summary for testphp.vulnweb.com (1–2 sentences):

This site is clearly a purposely vulnerable test application hosted on AWS with minimal security protections. It exposes things like directory listings, lacks HTTPS, and is designed to show common web vulnerabilities.

Summary for fullstackacademy.com (1–2 sentences):

Fullstack Academy has a normal public web presence, but there's still a lot of information available through Google like PDFs, syllabi, and subdomains. This kind of exposure can give insight into their structure and training content.

Top 3–5 Passive Findings:

Finding (what you discovered)	Tool / Query used	Why it matters to a defender
Directory listing exposed on vulnweb	Google dork (site:testphp.vulnweb.com)	This allows anyone to view files and directories, which could expose sensitive scripts or vulnerabilities.
Public PDFs and training documents on subdomains	site:start.fullstackacademy.com filetype:pdf	Documents can reveal internal structure, tools, or training content that could be useful for OSINT or targeting.
Public course syllabi available	site:fullstackacademy.com "syllabus"	Shows detailed info about technologies and topics being taught, which could help someone understand systems or focus areas.

Recommended follow-up action for a defender (1–2 sentences):

Overall, a defender should review what information is publicly exposed across both sites, including subdomains, documents, and directory access, and limit anything that doesn't need to be public. They should also make sure basic security controls like HTTPS and email protections are in place to reduce unnecessary risk and exposure.

Part 2 – Active Reconnaissance / Scanning (Metasploitable Lab)

Task 7 – Confirm Connectivity (Kali → Metasploitable)

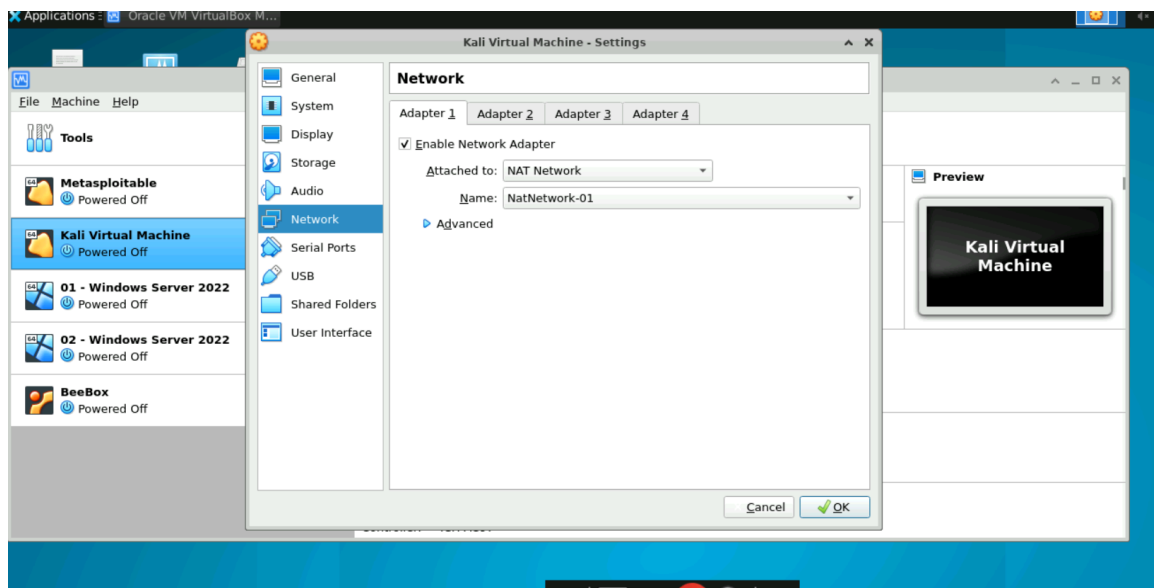
Instructions: Ensure your Kali system can reach the Metasploitable VM and record the details below.

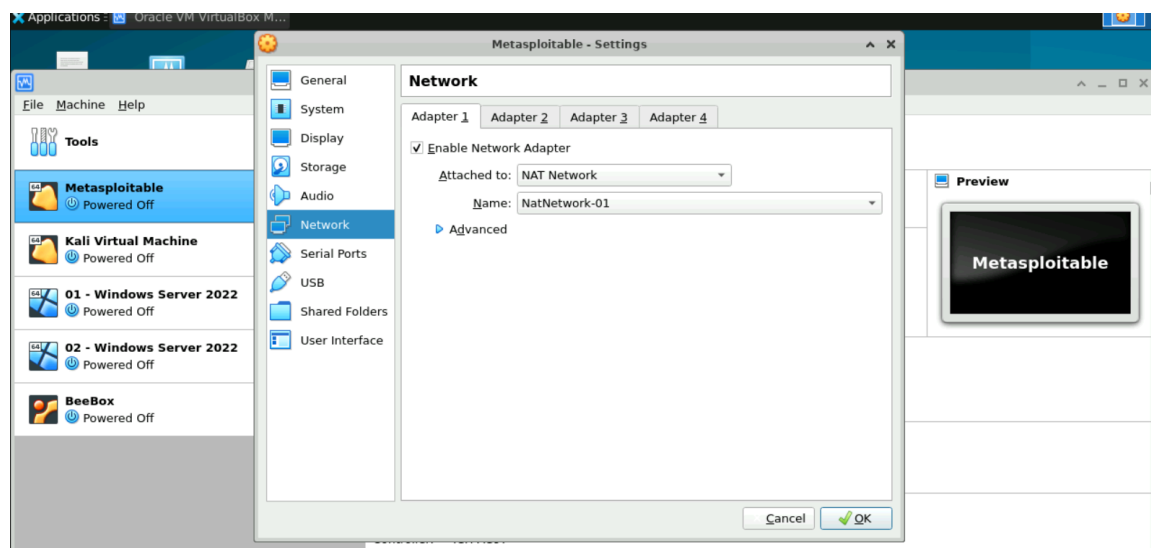
Metasploitable IP address: 10.0.2.8

Ping command used: ping 10.0.2.8

Result summary (e.g., number of successful replies): Ping worked and I was getting consistent replies back from 10.0.2.8 with no packet loss, so the connection between Kali and Metasploitable is good.

Screenshot filename(s):





Metasploitable IP 10.0.2.8

```
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To access official Ubuntu documentation, please visit:  
http://help.ubuntu.com/  
No mail.  
msfadmin@metasploitable:~$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000  
    link/ether 08:00:27:19:02:fb brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.8/24 brd 10.0.2.255 scope global eth0  
    inet6 fe80::a00:27ff:fe19:2fb/64 scope link  
        valid_lft forever preferred_lft forever  
3: eth1: <BROADCAST,MULTICAST> mtu 1500 qdisc noop qlen 1000  
    link/ether 08:00:27:39:b7:07 brd ff:ff:ff:ff:ff:ff  
msfadmin@metasploitable:~$ _
```

Kali IP - 10.0.2.5

```
labuser@kali: ~  
File Actions Edit View Help  
labuser@kali ~  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default  
    link/ether 08:00:27:b8:9b:d0 brd ff:ff:ff:ff:ff:ff  
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default  
    link/ether 08:00:27:d0:e6:d4 brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.5/24 brd 10.0.2.255 scope global dynamic noprefixroute eth1  
        valid_lft 407sec preferred_lft 407sec  
    inet6 fe80::a00:27ff:fed0:e6d4/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default  
    link/ether 02:42:04:6e:a9:30 brd ff:ff:ff:ff:ff:ff  
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0  
        valid_lft forever preferred_lft forever  
labuser@kali ~  
$
```


Ping from Kali to Metasploitable

```
labuser@kali:~$ ping 10.0.2.8
PING 10.0.2.8 (10.0.2.8) 56(84) bytes of data:
64 bytes from 10.0.2.8: icmp_seq=1 ttl=64 time=2.26 ms
64 bytes from 10.0.2.8: icmp_seq=2 ttl=64 time=0.741 ms
64 bytes from 10.0.2.8: icmp_seq=3 ttl=64 time=0.618 ms
64 bytes from 10.0.2.8: icmp_seq=4 ttl=64 time=0.785 ms
64 bytes from 10.0.2.8: icmp_seq=5 ttl=64 time=0.933 ms
64 bytes from 10.0.2.8: icmp_seq=6 ttl=64 time=0.477 ms
64 bytes from 10.0.2.8: icmp_seq=7 ttl=64 time=0.603 ms
64 bytes from 10.0.2.8: icmp_seq=8 ttl=64 time=0.823 ms
64 bytes from 10.0.2.8: icmp_seq=9 ttl=64 time=0.818 ms
64 bytes from 10.0.2.8: icmp_seq=10 ttl=64 time=1.21 ms
64 bytes from 10.0.2.8: icmp_seq=11 ttl=64 time=0.658 ms
64 bytes from 10.0.2.8: icmp_seq=12 ttl=64 time=0.987 ms
64 bytes from 10.0.2.8: icmp_seq=13 ttl=64 time=1.19 ms
64 bytes from 10.0.2.8: icmp_seq=14 ttl=64 time=0.871 ms
64 bytes from 10.0.2.8: icmp_seq=15 ttl=64 time=1.15 ms
64 bytes from 10.0.2.8: icmp_seq=16 ttl=64 time=0.617 ms
64 bytes from 10.0.2.8: icmp_seq=17 ttl=64 time=0.687 ms
64 bytes from 10.0.2.8: icmp_seq=18 ttl=64 time=0.669 ms
64 bytes from 10.0.2.8: icmp_seq=19 ttl=64 time=0.915 ms
64 bytes from 10.0.2.8: icmp_seq=20 ttl=64 time=0.963 ms
64 bytes from 10.0.2.8: icmp_seq=21 ttl=64 time=0.783 ms
64 bytes from 10.0.2.8: icmp_seq=22 ttl=64 time=0.872 ms
64 bytes from 10.0.2.8: icmp_seq=23 ttl=64 time=1.15 ms
64 bytes from 10.0.2.8: icmp_seq=24 ttl=64 time=1.39 ms
64 bytes from 10.0.2.8: icmp_seq=25 ttl=64 time=1.48 ms
64 bytes from 10.0.2.8: icmp_seq=26 ttl=64 time=1.69 ms
64 bytes from 10.0.2.8: icmp_seq=27 ttl=64 time=0.655 ms
64 bytes from 10.0.2.8: icmp_seq=28 ttl=64 time=0.691 ms
64 bytes from 10.0.2.8: icmp_seq=29 ttl=64 time=1.08 ms
64 bytes from 10.0.2.8: icmp_seq=30 ttl=64 time=0.530 ms
64 bytes from 10.0.2.8: icmp_seq=31 ttl=64 time=0.817 ms
64 bytes from 10.0.2.8: icmp_seq=32 ttl=64 time=0.518 ms
64 bytes from 10.0.2.8: icmp_seq=33 ttl=64 time=0.879 ms
64 bytes from 10.0.2.8: icmp_seq=34 ttl=64 time=0.690 ms
64 bytes from 10.0.2.8: icmp_seq=35 ttl=64 time=0.928 ms
64 bytes from 10.0.2.8: icmp_seq=36 ttl=64 time=0.600 ms
64 bytes from 10.0.2.8: icmp_seq=37 ttl=64 time=0.833 ms
64 bytes from 10.0.2.8: icmp_seq=38 ttl=64 time=0.603 ms
64 bytes from 10.0.2.8: icmp_seq=39 ttl=64 time=0.994 ms
64 bytes from 10.0.2.8: icmp_seq=40 ttl=64 time=0.915 ms
64 bytes from 10.0.2.8: icmp_seq=41 ttl=64 time=1.07 ms
64 bytes from 10.0.2.8: icmp_seq=42 ttl=64 time=0.698 ms
64 bytes from 10.0.2.8: icmp_seq=43 ttl=64 time=0.944 ms
64 bytes from 10.0.2.8: icmp_seq=44 ttl=64 time=0.934 ms
64 bytes from 10.0.2.8: icmp_seq=45 ttl=64 time=0.624 ms
64 bytes from 10.0.2.8: icmp_seq=46 ttl=64 time=0.856 ms
64 bytes from 10.0.2.8: icmp_seq=47 ttl=64 time=0.901 ms
64 bytes from 10.0.2.8: icmp_seq=48 ttl=64 time=0.759 ms
64 bytes from 10.0.2.8: icmp_seq=49 ttl=64 time=1.09 ms
64 bytes from 10.0.2.8: icmp_seq=50 ttl=64 time=0.673 ms
64 bytes from 10.0.2.8: icmp_seq=51 ttl=64 time=1.50 ms
64 bytes from 10.0.2.8: icmp_seq=52 ttl=64 time=0.840 ms
64 bytes from 10.0.2.8: icmp_seq=53 ttl=64 time=0.849 ms
64 bytes from 10.0.2.8: icmp_seq=54 ttl=64 time=0.827 ms
64 bytes from 10.0.2.8: icmp_seq=55 ttl=64 time=0.875 ms
64 bytes from 10.0.2.8: icmp_seq=56 ttl=64 time=0.794 ms
64 bytes from 10.0.2.8: icmp_seq=57 ttl=64 time=1.57 ms
64 bytes from 10.0.2.8: icmp_seq=58 ttl=64 time=1.34 ms
64 bytes from 10.0.2.8: icmp_seq=59 ttl=64 time=0.610 ms
64 bytes from 10.0.2.8: icmp_seq=60 ttl=64 time=0.983 ms
64 bytes from 10.0.2.8: icmp_seq=61 ttl=64 time=0.859 ms
```

```
File Actions Edit View Help
64 bytes from 10.0.2.8: icmp_seq=50 ttl=64 time=0.673 ms
64 bytes from 10.0.2.8: icmp_seq=51 ttl=64 time=1.50 ms
64 bytes from 10.0.2.8: icmp_seq=52 ttl=64 time=0.840 ms
64 bytes from 10.0.2.8: icmp_seq=53 ttl=64 time=0.849 ms
64 bytes from 10.0.2.8: icmp_seq=54 ttl=64 time=0.827 ms
64 bytes from 10.0.2.8: icmp_seq=55 ttl=64 time=0.875 ms
64 bytes from 10.0.2.8: icmp_seq=56 ttl=64 time=0.794 ms
64 bytes from 10.0.2.8: icmp_seq=57 ttl=64 time=1.57 ms
64 bytes from 10.0.2.8: icmp_seq=58 ttl=64 time=1.34 ms

64 bytes from 10.0.2.8: icmp_seq=59 ttl=64 time=0.610 ms
64 bytes from 10.0.2.8: icmp_seq=60 ttl=64 time=0.983 ms
64 bytes from 10.0.2.8: icmp_seq=61 ttl=64 time=0.859 ms
64 bytes from 10.0.2.8: icmp_seq=62 ttl=64 time=0.799 ms
64 bytes from 10.0.2.8: icmp_seq=63 ttl=64 time=0.942 ms
64 bytes from 10.0.2.8: icmp_seq=64 ttl=64 time=0.650 ms
64 bytes from 10.0.2.8: icmp_seq=65 ttl=64 time=0.721 ms
64 bytes from 10.0.2.8: icmp_seq=66 ttl=64 time=0.890 ms
64 bytes from 10.0.2.8: icmp_seq=67 ttl=64 time=0.729 ms
64 bytes from 10.0.2.8: icmp_seq=68 ttl=64 time=0.625 ms
^C
— 10.0.2.8 ping statistics —
68 packets transmitted, 68 received, 0% packet loss, time 67459ms
rtt min/avg/max/mdev = 0.477/0.897/2.256/0.303 ms

labuser@kali:~$
```

Task 8 – Basic Nmap Scan of Metasploitable

Instructions: Run a basic nmap scan against the Metasploitable IP address.

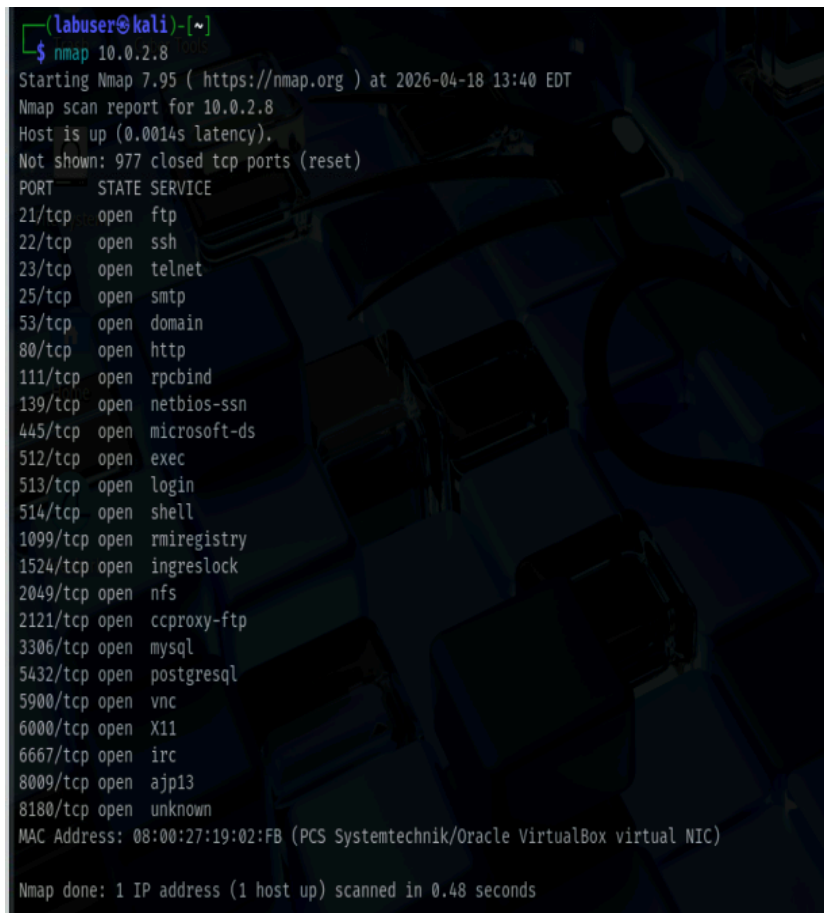
Nmap command used: `nmap 10.0.2.8 // nmap -sV 10.0.2.8`

General observations about the scan (number of open ports, anything that stands out):

The scan shows over 20 open ports, which is a lot and not typical for a secure system. It has multiple insecure services like FTP, Telnet, and SMB exposed, along with databases, which makes it clear this machine is intentionally vulnerable. Also MAC address for system is 08:00:27:19:02:FB

Screenshot filename(s):

`nmap 10.0.2.8`



```
labuser@kali:~$ nmap 10.0.2.8
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-18 13:40 EDT
Nmap scan report for 10.0.2.8
Host is up (0.0014s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:19:02:FB (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.48 seconds
```

nmap -sV 10.0.2.8

```

$ nmap -sV 10.0.2.8
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-18 13:45 EDT
Nmap scan report for 10.0.2.8
Host is up (0.00051s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:19:02:FB (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.13 seconds
```

Task 9 – Port & Service Interpretation

Instructions: Choose at least five open ports from your nmap results. For each, describe the service, a potential risk, and a simple defense idea.

Port	Service	Version	Example Risk	Simple Defense
21/tcp	FTP	vsftpd 2.3.4	This version is known to have a backdoor, so someone could get unauthorized access.	Disable FTP or upgrade to a secure version and use SFTP instead.
23/tcp	Telnet	Linux telnetd	Telnet sends data in plain text, so credentials can be easily intercepted.	Disable Telnet and use SSH instead.
80/tcp	HTTP	Apache httpd 2.2.8	Older web servers can have known vulnerabilities that attackers can exploit.	Keep the server updated and use HTTPS.
139/tcp	NetBIOS	Samba smbd 3.x	Samba services can expose file shares and be targeted for remote exploits.	Restrict access and keep Samba patched.
3306/tcp	MySQL	MySQL 5.0.51a	An exposed database could allow unauthorized access to sensitive data.	Restrict access to localhost and use strong credentials.

Task 10 – Active Recon Summary

Instructions: In 5–7 sentences, summarize what your scan results say about the Metasploitable system's exposure and what a defender should prioritize.

Summary:

The scan shows that the Metasploitable system is really exposed with a lot of open ports and services running. A lot of them are outdated or just not secure at all, like FTP and Telnet, which makes it pretty easy for someone to get in. Since the scan also shows the versions, someone could easily look up known vulnerabilities and take advantage of them. There are also bigger risks like databases being open and even a root shell, which is a huge red flag. Overall, it's clear this system isn't secured and is meant to be vulnerable.

From a defender side, the main focus should be closing unnecessary ports and getting rid of insecure services like Telnet and FTP. They should also update anything outdated and lock down access to things like databases. Adding basic security like firewalls, stronger login protection, and using encrypted connections would help a lot.